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Food choice and pharyngeal pumping activity of bacterial-feeding nematodes are driven by different functional traits

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ABSTRACT

The feeding behavior of free-living nematodes is shaped by several factors, including food choice and food consumption. The latter is commonly investigated by determining the pumping rate of the pharynx. However, whether the pharyngeal pumping activity also reflects food selection, i.e. whether the pumping rate for preferred food is increased, is largely unknown. The aim of this study was to investigate the relation between pumping rate and food choice. For this, the preference of the nematodes *Diploscapter coronatus*, *Diploscapter pachys*, *Plectus similis* and *Plectus* sp. were examined in binary choice assays. The bacteria *Bacillus subtilis*, *Kitasatospora* sp., *Sphingomonas trueperi*, *Pseudomonas fluorescens* and two *Legionella pneumophila* strains were tested against *Escherichia coli* OP50. The nematode pumping rates were determined for each bacterial diet. Additionally, the effects of bacterial species and size, and of nematode species and genus, size of buccal cavity and pharynx morphology, were investigated.

The nematodes showed a preference for *P. fluorescens* (except *D. pachys*) and avoidance of *B. subtilis*, *Kitasatospora* sp. and *L. pneumophila*. The determinants of food choice were nematode species, bacterial species and size, while the pumping rates varied with genus and pharynx proportion. The food preference determined by the choice assay was not necessarily associated with an upregulation of pharyngeal pumping, e.g. *Plectus* preferred *P. fluorescens* over *E. coli* OP50 when having a choice but pumped significantly faster when feeding on the latter. Although the pumping rate changed with bacterial species, there was no clear pattern that allowed a correlation with food choice across the tested nematode taxa, suggesting that pumping frequency is no indicator for food preference. Relating these results to nematode foraging strategy, food choice was determined by bacterial traits associated with diet quality and search, while food consumption was linked to nematode traits affecting diet handling.

1. Introduction

Food intake is essential for the survival for every animal, and is regulated by external environmental and internal physiological factors (Tierney, 2020). Key determinants include the animals' nutritional requirements, satiety state, reproductive state, food composition, presence of toxic chemicals and predation risk (de Carvalho and Mirth, 2017). According to foraging theory, foraging animals make trade-offs between these different aspects when foraging, in order to maximize their energy intake per time (Stephens and Krebs, 1986).

Bacterial-feeding soil nematodes, despite consuming a wide range of bacterial diets, generally select those that are favorable for growth and reproduction (Newsham et al., 2004; Salinas et al., 2007). This selective feeding significantly effects longevity and egg production (Venette and Ferris, 1998) and can be adjusted to physiological needs. Under stress, *Caenorhabditis elegans* shifts its diet from food that supports reproduction and a shortened lifespan to food that maintains survival but reduces brood size (Liu et al., 2021).

The metabolic and feeding activities of nematodes shape their functional role within the soil food web (Ferris, 2010; Yeates, 2010). Especially, microbial grazers are crucial for mineralization processes and soil nutrient cycling (Ruess, 2024). For bacterial-feeding nematodes the global abundance was assigned as high as 1.9×10^{20} Ind g^{-1} soil dry weight (van den Hoogen et al., 2019). Typically, they take up more

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nitrogen (N) than is needed for their body functions, resulting in excess N excreted as ammonia available for plant uptake, which is approximately 32–38 % of the annual N mineralization in arable land (Whalen et al., 2013). Further, these trophic interactions directly affect microbial activity and carbon (C) release from microbial biomass (Richter et al., 2019). Bacterial grazers are also main drivers of the soil microbial loop, a tri-trophic interaction rendering nutrients available to plants (Bonkowski et al., 2009; Zheng et al., 2022). A crucial component of consumer-prey models is the concept of functional response, which relates the number of food items consumed per time unit to food abundance (Jeschke and Hohberg, 2008). For the soil microfauna, studies on functional responses are available for predators such as Tardigrades (Hohberg and Traunspurger, 2005) as well as filter feeders like protozoa (Jeschke et al., 2004), while the knowledge on nematodes is minor. Functional responses are mainly affected by search time (i.e. time to find a food item), handling time (i.e. time to attack and eat a food item) and digestion time (i.e. gut transit time of food item) of the consumer (Jeschke et al., 2004). For example, handling time in bacterial-feeding nematodes is linked to the morphology of the buccal cavity, e.g. funnel-shape (Cephalobidae) or rod-shape (Rhabditidae) affect capturing of bacterial diets as well as size of ingested cells, with smaller bacteria more easily swallowed (Avery and Shtonda, 2003).

Nematodes use olfactory, mechanical and chemical cues to detect their bacterial prey (Meisel and Kim, 2014; Kim and Flavell, 2020). While foraging, nematodes switch between two discrete foraging states called roaming and dwelling (Flavell et al., 2013). Dwelling animals show frequent turns and reversals at a low speed in a small area, while roaming individuals move rapidly with only few reversals to explore the environment (Ben Arous et al., 2009). Nematodes modulate their foraging activity depending on the food quality, i.e. the time spent roaming increases with decreasing nutritional value of the diet (Shtonda and Avery, 2006).

Foraging and food preference of nematodes are traditionally studied in food choice experiments, in which animals two or more diet options are offered (Shtonda and Avery, 2006; Abada et al., 2009; Liu et al., 2017). Such bioassays revealed that traits of the bacterial prey (Salinas et al., 2007; Lewenza et al., 2014; Yu et al., 2015; Liu et al., 2018), but also exposure to toxicants (Monteiro et al., 2014; Pacheco et al., 2022) shape the food choice of nematodes: Typically, small-sized, Gram--negative bacteria that fit the stoma are preferred (Rønn et al., 2012; Yu et al., 2015) as well as cells with a high water or nutrient content (Yu et al., 2015; Liu et al., 2017). On the other hand, nematodes avoid bacteria producing nematicidal metabolites such as chitinases, proteases or hydrogen cyanide (Berini et al., 2018; Khan et al., 2018).

Feeding in nematodes is closely linked to the pumping activity of the pharynx, a neuromuscular tube transporting ingested bacteria from the buccal cavity to the intestine (Altun and Hall, 2009; Luedtke et al., 2010). In the absence of food, the pharynx of *C. elegans* pumps once every second, whereas in the presence of food the pumping rate is upregulated to around four pumps per second (Franks et al., 2006). Pumping rate is also affected by satiety state, i.e. in the presence of food a starving nematode will increase the pumping rate faster than a well-fed individual (Avery et al., 1993). Moreover, life strategy may play a role: With food available, enrichment opportunists such as Rhabditidae show an almost constant ingestion of the microbial suspension, whereas in general opportunists such as Cephalobidae or Plectidae feeding on bacteria appears more deliberate (Bongers and Ferris, 1999).

In *C. elegans* as well as other rhabditids e.g. *Diploscapter coronatus*, the pharynx is divided into three functional parts: The corpus, isthmus and terminal bulb (Avery and Shtonda, 2003). The pharynx shows two distinct muscle movements, the pharyngeal pumping and isthmus peristalsis (Song and Avery, 2013). Pumping draws bacteria into the pharyngeal lumen, while isthmus peristalsis transports bacteria to the terminal bulb (Fang-Yen et al., 2009). Other types exist such as the less complex two-part pharynx of the genus *Plectus* lacking a metacorpus and isthmus. Here, the pharynx consists of a muscular cylinder of equal width and a terminal bulb called propharynx and a terminal bulb (von Lieven, 2003). This evolutionary diversity in morphology has implications for the pharyngeal physiology of taxa, such as muscle activity (Tahseen et al., 2004) or the pattern of pumping and peristalsis (Chiang et al., 2006).

Pharyngeal pumping assays are mainly used to examine the therapeutic effects of pharmaceutical drugs (Aitlhadj and Stürzenbaum, 2013; Diomede et al., 2014; Gaur and Agarwal, 2021; Izquierdo et al., 2023; Pereira et al., 2024), but also in fatty acid metabolism research (Zhang et al., 2011; 2015). Reports correlating pharyngeal pumping activity with food choice are scarce and restricted to the model nematode *C. elegans* (Abada et al., 2009). Comparing the pumping rates of *C. elegans* when fed with either *E. coli* or artificial particles revealed that the nutritional value of a diet (i.e. its food quality) is essential to upregulate the pumping activity (Fueser et al., 2021). Thus, we hypothesize that the preference for a bacterial species will be reflected by an increase in the pharyngeal pumping frequency of common bacterial-feeding nematodes in the environment.

Experiments were performed with the widespread soil bacteria *Bacillus subtilis*, *Kitasatospora* sp., *Sphingomonas trueperi*, *Pseudomonas fluorescens,* as well as two strains of the human pathogen *Legionella pneumophila.* Food preferences were compared to the standard lab food *Escherichia coli* OP50. To account for different life-history traits and pharynx morphology, the nematode species *Diploscapter coronatus* and *Diploscapter pachys* (three-part pharynx, enrichment opportunist), as well as *Plectus similis* and *Plectus* sp. (two-part pharynx, general opportunist) were chosen for experiments. Binary food choice tests and pharyngeal pumping assays were conducted, and the nematode food preferences indicated by the two methods are compared. The potential of pumping assays as alternative test approach is discussed.

2. Materials and methods

2.1. Nematode grazers

The nematode species used for the experiments were the bacterial feeders *Diploscapter coronatus* (Cobb 1893), *Diploscapter pachys* (Steiner 1942), *Plectus similis* (Zell, 1993) and *Plectus* sp. In the environment, the thermophile *D. coronatus* is abundant in compost, sewage and the rhizosphere of arable soil (Gibbs et al., 2005). The thermophile *D. pachys* is also frequent in soil but is less slender and moving slower than *D. coronatus* (pers. observation). *P. similis* can be found in soil, moss, litter and biofilms (Zell, 1993; Hemmerling et al., 2023), while *Plectus* sp. occurs in a thermal spa biofilm (Rasch et al., 2016).

For the current study, *D. coronatus* was kindly provided by Dr. Philipp Schiffer and *D. pachys* strain PF1309 by Hélène Fradin (Fradin et al., 2017). Stock cultures of both species were grown on nematode growth medium (NGM) agar plates inoculated with an *Escherichia coli* OP50 lawn at 20 °C. *P. similis* was isolated from a cooling tower biofilm in Lower Saxony, Germany (Hemmerling et al., 2023). *Plectus* sp. was isolated from a biofilm from a thermal spa bath in Aix-les Bains, France (Rasch et al., 2016). Both species were grown on Page's Amoeba saline (PAS) agar without the addition of food. Nematode body length and pharyngeal dimensions were determined microscopically (Leica TCS SP8, Wetzlar, Germany) for each tested nematodes species (each species n = 15 females). The width of the buccal cavity was measured at the cheilostom, i.e. at the most anterior part. The pharynx length equals the distance from the posterior end of the stoma to the posterior end of the terminal bulb, while the pharynx proportion refers to the body length divided by pharynx length.

2.2. Bacterial diet

The study was carried out with a range of bacteria present in environmental samples (Fig. S1, Tab. S1). Common soil bacteria were the Gram-positive *Bacillus subtilis* (DSM 10, Firmicutes) and *Kitasatospora* sp.

(DSM 45065, Actinobacteria), as well as the Gram-negative *Sphingomonas trueperi* (DSM 7225, Proteobacteria) and *Pseudomonas fluorescens* (DSM 50090, Proteobacteria). All bacterial strains were provided by the Leibnitz-Institute DSMZ GmbH (German Collection of Microorganisms and Cell Cultures, Braunschweig, Germany). Further, two strains of the aquatic human pathogen *Legionella pneumophila* (Proteobacteria) were tested: (i) the environmental isolate KV02 isolated from a cooling tower (for details see Hemmerling et al., 2023) and (ii) the strain Corby (Wintermeyer et al., 1995). The lab strain *Escherichia coli* OP50 (Proteobacteria) served as control.

B. subtilis, *Kitasatospora* sp., *S. trueperi* and *P. fluorescens* were grown in Standard Nutrient Broth I (Carl Roth, Karlsruhe, Germany). *E. coli* was cultured in lysogeny broth (LB broth). *B. subtilis* and *E. coli* were incubated for 16 h at 37 °C. *P. fluorescens*, *S. trueperi* and *Kitasatospora* sp. were incubated at 20 °C for 48 h. *L. pneumophila* was grown on buffered charcoal-yeast extract (BCYE) agar at 37°C for 48 h. After incubation, bacteria were harvested and culture media/agar washed of two times. Each washing step comprised centrifugation (3000 g, 15 min), discarding of the supernatant and floating with phosphate-buffered saline (PBS). A bacterial concentration of 10^6 colony forming units (CFU) ml^{-1} PBS was used for all experiments. The concentration was determined via serial dilution and plate counting. Bacterial sizes were determined microscopically (Fig. S1) (VHX-600 digital microscope; Keyence Corporation, Osaka, Japan).

2.3. Food choice assay

All binary food choice assays were performed on 9 cm NGM agar plates at 20°C. In each trial, the lab strain *E. coli* OP50 was tested against one of the six test bacteria. For this, 30 μl of freshly prepared bacterial suspensions adjusted to 10^6 CFU ml^{-1} were seeded at opposite sides of the assay plate at a distance of 1.5 cm from the plates' edge. The dried suspensions formed bacterial lawns with a diameter of 1 cm. In a next step, 100 young adult nematodes suspended in sterile mineral water (Volvic, Danone Deutschland GmbH, Germany) were placed at the center of each plate. Prior to inoculation, the nematodes were subjected to a washing procedure comprising two washing steps with sterile novobiocin solution (40 mg l^{-1} Volvic) for 10 min each, followed by two washing steps with sterile Volvic. After 8, 24 and 48 h the number of individuals that moved to the bacterial lawns were counted and the Choice Index (CI) after Abada et al. (2009) was calculated as followed: $CI = TB - CB / T$, where *TB* represents the number of individuals in the test bacterium, *CB* the number of individuals in the control bacterium *E. coli* OP50, and *T* the total number of tested individuals. A CI of −1.0 represents a total preference for *E. coli* OP50, a CI of 1.0 represents a total preference for the test bacterium, while a CI of 0 represents an equal distribution of nematode individuals between both bacterial diets.

Each trial was replicated 5 times with 3 repeats per replicate (a total of 15 assay plates per test bacterium and nematode species). Additionally, a control assay with sole PBS buffer at both ends of the assay plate verified that nematodes did not move in a directed manner towards the buffer. The control assay was repeated 5 times for each nematode species (50 individuals per replicate) and the number of nematodes in the PBS spots was counted after 8, 24 and 48 h.

A series of preliminary tests was conducted to determine the optimal duration of the food choice assays. This step was necessary as the tested nematode species differ in their individual foraging behavior. In detail, *E. coli* OP50 was tested against *P. fluorescens* (considered as an attractive diet) to trigger a strong food choice. The preliminary tests were repeated 5 times for each nematode species (40 individuals per replicate) and the food choice was observed 1 h, 2 h, 4 h, 6 h, 8 h, 24 h and 48 h after bacterial application (a total of 20 individual assays). The time at which all four nematode species showed a clear choice was chosen for the assays.

2.4. Pharyngeal pumping assay

Prior to the pharyngeal pumping assays, the optimal adaption time, i.e. the time at which the highest proportion of individuals was actively pumping with their pharynx was determined. For each nematode species, bacteria suspensions were fed to 60 specimen (3 replicates with each 20 individuals) and their feeding behavior was monitored over a period of 8 hours (for details see Hemmerling et al., 2023). This revealed the following optimal experimental time frames: For *P. similis* and *Plectus* sp., adaption time was 1 h for *Kitasatospora* sp., 3 h for *P. fluorescens*, *S. trueperi*, *L. pneumophila* KV02 and Corby, and 8 h for *B. subtilis* and *E. coli* OP50. For *D. coronatus* and *D. pachys* the adaption time was 5 h for all tested bacterial strains.

For the main pumping assay, 300 μl of a freshly prepared bacteria suspension were seeded onto a NGM agar plate and allowed to dry. Then, 100 young adult nematodes were hand-picked from a culture plate onto the bacterial lawn (3 replicate plates with n = 100 individuals). After the respective adaption time for the respective binary link, the pharyngeal pumping rate of the nematodes was determined. For this, 1 min videos of individuals feeding (500x magnification) were recorded using the VHX-600 digital microscope (Video S1). The videos were then played back at half speed to count each individual pump. On each agar plate videos of 10 different individuals were recorded and analyzed.

2.5. Statistics

All statistical analyses were performed using the software R version 4. 3. 2 "Eye holes" (R Core Team, 2023) and the packages "car" (Fox and Weisberg, 2019), "rstatix" (Kassambara, 2023), "MASS" (Venables and Ripley, 2002) and "lmerTest" (Kuznetsova et al., 2017). Prior to statistical tests, data sets were checked for potential outliers with observations outside the 5 and 95 percentiles being discarded as outliers. Data which did not meet the assumptions for normal distribution and homogeneity of variances, were transformed using the Box Cox transformation or the arcsine transformation (CI values only). Then, the data were either subjected to a one-way analysis of variances (ANOVA) followed by the Tukey HSD post hoc test (pumping rates), or a repeated measures ANOVA followed by a multiple pairwise comparisons as post hoc test (CI values). A significance level of $p < 0.05$ was applied to all statistical analyses.

The impact of different nematode and bacterial traits on the CI and pharyngeal pumping frequency was analyzed using linear mixed effect models (LMMs). For the CI model the factors "time", "nematode species" (to account for species-specific food preferences), "width of buccal cavity", "bacterial species" and "bacterial size" were set as fixed effects. For analyzing the pharyngeal pumping frequency, the factors "pharynx proportion", "nematode genus" (to account for taxon-specific pharynx morphology), "bacterial species" and "bacterial size" were set as fixed effects. The factor "Replicate" was set as random effect in both models.

3. Results

3.1. Morphology of buccal cavity and pharynx

The buccal cavity at the cheilostom showed species-specific width differences ($p < 0.001$; Fig. 1). *D. pachys* had the widest stoma (3.5 μm), followed by *D. coronatus* (3.2 μm). The buccal cavity of *P. similis* (2.5 μm) was significantly narrower than that of both *Diploscapter* species ($p < 0.01$). With 1.9 μm width *Plectus* sp. had the smallest stoma of all four species ($p < 0.01$).

The pharynx length showed also distinct species-specific differences ($p < 0.001$; Table 1). *P. similis* had the longest pharynx with 120 μm ($p < 0.001$), followed by *Plectus* sp. with 95 μm. Both *Diploscapter* species had a significantly shorter pharynx than *Plectus* ($p < 0.05$) with 83 and 77 μm for *D. coronatus* and *D. pachys*, respectively. On the other hand, the

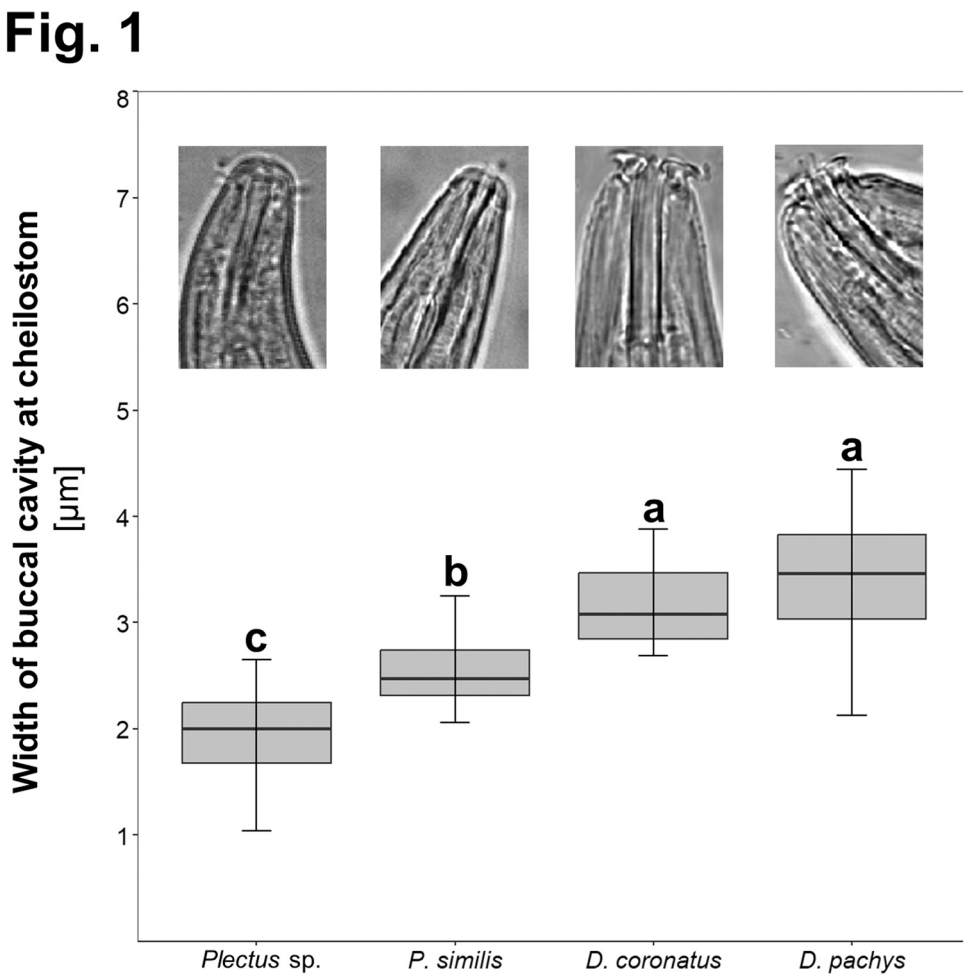

Fig. 1. Width of buccal cavity at cheilostom [µm] of the nematodes *Diploscapter coronatus*, *Diploscapter pachys, Plectus similis* and *Plectus* sp. (each species n = 15 females). Boxplots: The median is shown by a solid line. The boxes represent the interquartile range (i.e. 50 % of the data) and the whiskers represent the data that are in the range of 1.5 times the interquartile range. Statistical differences according to Tukey's HSD test at $p < 0.05$.

Table 1
Length [µm] of pharyngeal body sections of the free-living nematodes *Diploscapter coronatus*, *Diploscapter pachys*, *Plectus similis* and *Plectus* sp. (each species n = 15 females). Colonizer (*c*) and persister (*p*) values after Bongers (1990). Statistical differences according to Tukey's HSD test or *t*-test at $p < 0.05$.

	c* – *p	**Total body length**	**Pharynx**					**Grinder**	**Pharynx/body length** [%]
			Procorpus	Metacorpus	Isthmus	Terminal bulb	Total		
D. coronatus	1	445.3 ± 62.5 **b**	25.5 ± 3.6 **a**	14.9 ± 2.2	22.6 ± 1.7 **a**	20.1 ± 2.9 **a**	83.1 ± 3.9 **c**	Yes	18.9 ± 2.4 **b**
D. pachys	1	380.0 ± 61.3 **b**	21.8 ± 2.7 **b**	14.3 ± 1.9	19.5 ± 2.7 **b**	21.0 ± 3.8 **a**	76.6 ± 6.2 **c**	Yes	20.4 ± 2.6 **ab**
			Propharynx			Terminal bulb	Total		
P. similis	2	531.6 ± 104.5 **a**	99.9 ± 15.2 **a**			20.0 ± 1.8 **a**	119.9 ± 15.6 **a**	Yes	23.1 ± 3.4 **a**
Plectus sp.	2	435.8 ± 79.7 **b**	79.0 ± 9.7 **b**			16.1 ± 2.1 **b**	95.1 ± 10.9 **b**	Yes	22.3 ± 3.1 **a**

proportion of the pharynx length to the total body length differed significantly between *D. coronatus* (19 %) and *P. similis* (23 %; $p < 0.01$) and *Plectus sp.* (23 %; $p < 0.05$), respectively. Interestingly, *Plectus* sp. had the shortest terminal bulb of all four species ($p < 0.01$). Additionally, significant length differences occurred for the procorpus ($p < 0.01$) and isthmus ($p < 0.01$) between *D. coronatus* and *D. pachys*, as well as for the propharynx between *P. similis* and *Plectus* sp. ($p < 0.001$).

3.2. Choice experiment – food choice assay

In binary food choice assays with *E. coli* OP50 as control *D. coronatus* showed a strong preference for *P. fluorescens* with a CI of 0.32 after 8 h, which significantly increased to a CI of 0.6 after 48 h ($p < 0.001$; Fig. 2). *D. coronatus* exhibited a less pronounced preference for *S. trueperi* with CI values ranging from 0.13 after 8 h to 0.23 after 48 h ($p < 0.05$). When offered *L. pneumophila*, *B. subtilis* and *Kitasatospora* sp. as alternative diet, *D. coronatus* preferred *E. coli* OP50, and this preference increased significantly with time.

In *D. pachys* food preference were similar to *D. coronatus* (Fig. 2). In detail, an increasing choice for *E. coli* OP50 occurred with time, when compared to *L. pneumophila* KV02 ($p < 0.001$), *L. pneumophila* Corby ($p < 0.001$), *B. subtilis* ($p < 0.01$) and *Kitasatospora* sp. ($p < 0.001$).

P. similis exhibited a distinct food choice, with interaction strength depending on the respective binary link. A strong preference for *P. fluorescens* occurred, additionally increasing with time with CI values from 0.15 after 8 h, over 0.52 after 24 h ($p < 0.001$) to 0.77 after 48 h

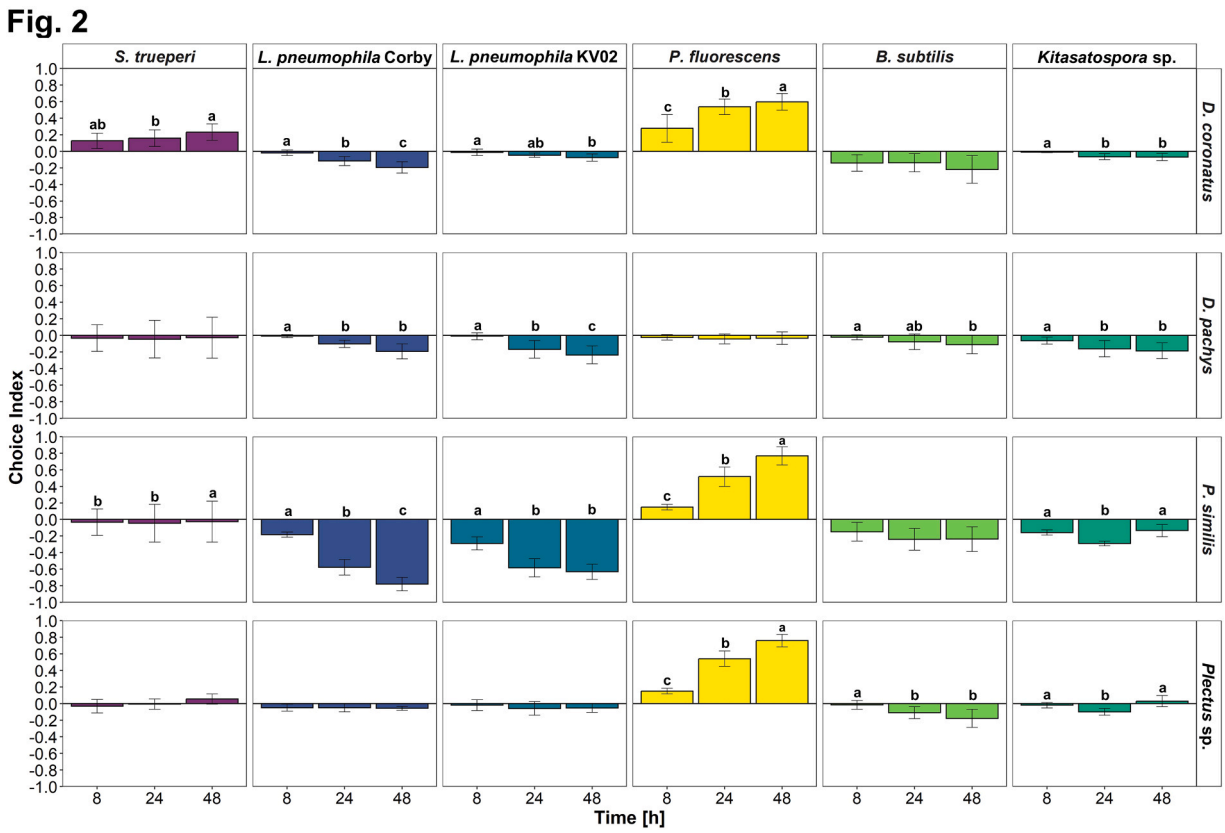

Fig. 2. The Choice Index (CI) of six different bacteria when compared to *Escherichia coli* OP50 for the nematodes *Diploscapter coronatus*, *Diploscapter pachys, Plectus similis* and *Plectus* sp. (5 replicates per nematode species, each with 3 observational trials). Bars with no or the same letter are not significantly different according to Tukey's HSD test at $p < 0.05$.

($p < 0.001$; Fig. 2). On the other hand, both *L. pneumophila* strains triggered a strong rejection response, which got stronger with time ($p < 0.001$). The CI for *L. pneumophila* KV02 decreased to −0.63 and for *L. pneumophila* Corby to −0.78 within 48 h. *P. similis* showed a weak preference for *E. coli* OP50, when *B. subtilis* was offered as alternative diet with CI value of −0.24 after 48 h. Interestingly, fluctuating CI values were observable for *Kitasatospora* sp.: After the CI initially decreased from −0.16 after 8 h to −0.29 after 24 h ($p < 0.001$), it increased again to a value of -0.13 ± 0.07 after 48 h ($p < 0.01$).

Similar to *P. similis, Plectus* sp. exhibited a strong preference for *P. fluorescens* (Fig. 2), and the CI increased with time from 0.15 (8 h) over 0.54 (24 h, $p < 0.001$) to 0.76 (48 h, $p < 0.001$). Also, *Plectus* sp. preferred *E. coli* OP50, when *B. subtilis* was offered. Moreover, this preference increased with time from a CI of −0.02 after 8 h, over −0.11 after 24 h ($p < 0.05$) to −0.18 after 48 h ($p < 0.01$). Interestingly, *Plectus* sp. showed a diet switch from when given a choice between *E. coli* OP50 and *Kitasatospora* sp., which was significant with a CI of −0.1 after 24 h significantly increasing to 0.03 after 48 h ($p < 0.001$).

3.3. No-choice experiment – pharyngeal pumping assay

The pumping rates differed with bacterial diet and nematode consumer (Fig. 3). The pumping activity of *D. coronatus* was highest when fed with *Legionella*, with 94 and 86 pumps min^{-1} for *L. pneumophila* Corby and *L. pneumophila* KV02, respectively ($p < 0.001$). Intermediate pumping rates were observed for *B. subtilis* (48 pumps min^{-1}) and *Kitasatospora* sp. (46 pumps min^{-1}). The pumping activity was lowest for *S. trueperi*, *P. fluorescens* and *E. coli* OP50 with pumping rates around 25 pumps min^{-1} ($p < 0.001$).

For *D. pachys*, similarly high pumping rates were observed when fed with *B. subtilis* (103 pumps min^{-1}) or *P. fluorescens* (98 pumps min^{-1}). The pumping rate differed between the *Legionella* strains and was higher for *L. pneumophila* Corby (98 pumps min^{-1}), than that for *L. pneumophila* KV02 (52 pumps min^{-1}; $p < 0.001$). The pumping rate for *E. coli* OP50

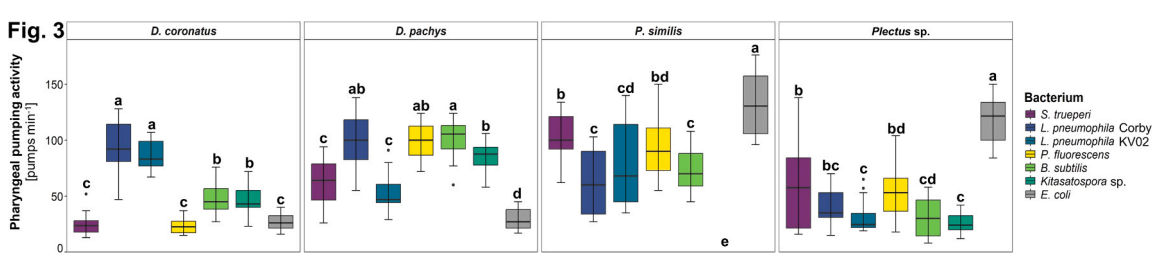

Fig. 3. Pharyngeal pumping activity per individual [contractions min^{-1}] of the nematodes *Diploscapter coronatus*, *Diploscapter pachys, Plectus similis* and *Plectus* sp. (3 replicates per nematode species) fed with seven different bacterial taxa. Boxplots: The median is shown by a solid line. The boxes represent the interquartile range (i. e. 50 % of the data) and the whiskers represent the data that are in the range of 1.5 times the interquartile range. Statistical differences according to Tukey's HSD test at $p < 0.05$.

with 29 pumps min^{-1} was lower ($p < 0.001$) than for all other bacterial diets.

Contrasting this, *E. coli* OP50 triggered the highest pumping activity in *P. similis* with 133 pumps min^{-1} ($p < 0.001$). Intermediate pumping rates were observed for *S. trueperi* (105 pumps min^{-1}) and *P. fluorescens* (96 pumps min^{-1}). Compared to *S. trueperi* and *P. fluorescens*, the pumping rate of *P. similis* was lower with *L. pneumophila* Corby ($p < 0.001$) and *B. subtilis* ($p < 0.05$). Surprisingly, no pumping activity could be detected in *P. similis* when *Kitasatospora* sp. was offered as diet.

Like *P. similis*, *Plectus* sp. pumped fastest when fed with *E. coli* OP50 (119 pumps min^{-1}; $p < 0.001$). The pumping activity for *S. trueperi* (60 pumps min^{-1}) and *P. fluorescens* (53 ± pumps min^{-1}) decreased by half. Otherwise, the pumping rates ranged between 26 and 40 pumps min^{-1} for *Kitasatospora* sp. and *L. pneumophila* Corby, respectively.

3.4. Effects of prey size and nematode morphology on food choice and ingestion

The results delineated above suggest that food preference is not correlated to food ingestion. For example, *P. similis* and *Plectus* sp. preferred *P. fluorescens* over the control *E. coli* OP50 in the food choice assay (Fig. 2), yet pharyngeal pumping rates were significantly higher in *E. coli* OP50 (Fig. 3).

To determine the factors that shaped the observed food choice and ingestion linear mixed effect models were applied. The food choice, i.e. the CI, was strongly affected by the bacterial prey species ($p < 0.001$) and cell size ($p < 0.001$), yet not by stoma width of the nematode consumer (Table 2). Additionally, food choice was influenced by nematode species ($p < 0.001$) but not by time. In contrast to food choice, bacterial prey species ($p = 0.120$) and cell size ($p = 0.998$) were irrelevant for food ingestion, i.e. pharyngeal pumping frequency (Table 3). Instead, the consumer morphology was important. Both, the pharynx proportion ($p < 0.001$) and genus (three-part or two-part pharynx; $p < 0.003$) affected the pumping frequency.

4. Discussion

4.1. Food choice of bacterial grazers

The feeding preference of bacterial-feeding nematodes generally is a response to the physico-chemical properties of their diet (Abada et al., 2009; Yu et al., 2015; Liu et al., 2017). The binary food choice assay with four bacterial-feeding nematode species revealed that all tested species (except *D. pachys*) strongly preferred *P. fluorescens*, likely due to small alcohols, esters and ketones released, functioning as chemo-attractants (Bargmann et al., 1993). Further, Yu et al. (2015) related the attraction to *P. fluorescens* to high concentrations of CO_2 released by respiration, as this generally indicates large and active bacterial populations (Rasmann et al., 2012). Moreover, *P. fluorescens* cells show a high water content, which can support nematode metabolic functions and maintenance of their osmolytic balance (Bongers and Ferris, 1999; Liu et al., 2017). The preference for *P. fluorescens* was also indicated by the number of laid eggs, which was clearly higher there than in *E. coli* OP50 (pers. observation).

Table 2
Linear mixed effect model table of the effect of time, cheilostom width and bacterial size and species on the food choice (Choice index) of the tested nematode species.

Factor	Mean of square	df1	df2	*F* value	*p* value
Time	0.179	2	345	1.984	0.139
Species	3.5	2	345	38.783	**< 0.001**
Cheilostom width	0.009	1	345	0.01	0.752
Bacterial species	9.942	4	345	110.164	**< 0.001**
Bacterial size	4.435	1	345	49.139	**< 0.001**

df1 numerator degree of freedom, df2 denominator degree of freedom. Significant effects are in bold.

Table 3
Linear mixed effect model table of the effect of bacterial and nematode morphology on the pharyngeal pumping frequency of the nematode genera *Diploscapter* and *Plectus*.

Factor	Mean of square	df1	df2	*F* value	*p* value
Genus	7821919	1	75	9.769	**0.003**
Pharynx proportion	10590686	1	75	13.227	**0.001**
Bacterial species	1453169	5	75	1.815	0.120
Bacterial size	7	1	75	0.000	0.998

df1 numerator degree of freedom, df2 denominator degree of freedom. Significant effects are in bold.

The tested nematode species responded differently to *S. trueperi*: Specimen either preferred (*D. coronatus*) or avoided (*P. similis*) *S. trueperi* or exhibited no clear choice (*D. pachys* and *Plectus* sp.) for either *S. trueperi* or *E. coli* OP50. This contrasts other studies, e.g. based on ^{13}C-labeling microcosm experiments Richter et al. (2019) suggested *S. trueperi* is a high-quality food for *C. elegans* and *Acrobeloides buetschlii*. In line with this, Stuhr and Curran (2020) reported that a bacterial diet containing *Sphingomonas* resulted in a faster development from the 4th larval stage (J4) to adulthood and an increased lifespan in *C. elegans* compared to *E. coli* OP50.

Both, *B. subtilis* and *Kitasatospora* sp. were avoided by all tested nematode species. Morphological defense mechanisms of these bacteria include biofilm and filament formation and the establishment of microcolonies, i.e. forming dense cell clusters that are too big for consumption (Matz and Kjelleberg, 2005). Additionally, both produce a variety of secondary metabolites with nematicidal properties: The spore-forming *B. subtilis* is one of the most studied biocontrol agents and provides effective control of plant-parasitic nematodes such as *Meloidogyne* spp., *Pratylenchus* spp. and *Rotylenchus reniformis* (Higaki and Araujo, 2012; Mazzuchelli et al., 2020). These antagonistic activities include the egg deformation and inhibition of egg hatching as well as the increase of J4 mortality, as shown for *M. incognita* (Rao et al., 2017). *Kitasatospora* produces bafilomycin B1, which expresses nematocidal activities against the plant-parasitic nematode *Bursaphelenchus lignicolus* (Otoguro et al., 1988). Moreover, exposure to different phenazine derivates synthesized by *Kitasatospora* increased protein misfolding of polyglutamine in nematode muscle cells in *C. elegans* and suppressed hatching of larvae in *M. incognita* (Kavitha et al., 2005; Ray et al., 2015).

L. pneumophila occurs ubiquitously in natural aquatic environments, where it parasitizes intracellularly mainly within protozoa (Molmeret et al., 2005). Recently, the environmental co-occurrence of *L. pneumophila* and free-living nematodes was reported from biofilms of a natural hot spring and several cooling towers (Rasch et al., 2016; Paranjape et al., 2020; Hemmerling et al., 2023). Infection studies with *C. elegans* revealed that *L. pneumophila* is able to colonize and persist in the intestinal tract of nematodes (Brassinga et al., 2010). Following the invasion of intestinal cells, *L. pneumophila* forms *Legionella*-containing vacuoles (LCVs), which contribute to the persistence and pathogenicity of the bacterial cells (Hellinga et al., 2015). Pathogen infection results in a shortened life span of *C. elegans*, accompanied by the extrusion of viscera through the vulva as well as intestinal and anal distensions (Brassinga et al., 2010). Taken together, these findings suggest that nematodes avoid *L. pneumophila* as food source, which was proven by the present experiments for all tested nematode species, when offering *E. coli* OP50 as alternative diet.

The food choice of nematodes was further related to bacterial size. For example, all tested nematode species (except *D. pachys*) preferred the smaller *P. fluorescens* (2.69 × 0.97 μm) over *E. coli* OP50 (2.87 × 1.26 μm). Further, *Kitasatospora* sp., which had the largest cell size (4.53 × 1.46 μm) was avoided by all nematodes. These findings are in line with other studies reporting a preference for nutrient-rich, small-sized bacteria (Salinas et al., 2007; Yu et al., 2015). Avery and Shtonda,

(2003) showed that nematode growth rates declined with increasing bacterial cell size and concluded that consumption is reduced for large cells. However, the preference for smaller bacteria was not consistent, as both strains of *Legionella* were avoided by all nematodes, although they had a smaller cell size compared to *E. coli* OP50. This is probably due to pathogenic nature of *L. pneumophila* (see above). Correspondingly, when choosing between cells of comparable size, i.e. *E. coli* OP50 and *B. subtilis*, all nematodes did avoid the latter with its known nematicidal properties.

The width of the buccal cavity was not relevant for choosing a potential prey. Moens et al. (2004) argued that stoma dimensions do not correctly reflect the size range of ingestible particles, as ingestion of cells much larger than the stoma diameter was reported for aquatic *Alaimus* and *Theristus*. Moreover, Moens and Vincx (1997) suggested that food choice resides at the level of digestion rather than ingestion, as food particles can be defecated in undigested conditions. Next to the discussed defense mechanisms, this may further explain the observed avoidance of *B. subtilis* and *Kitasatospora* sp. As Gram-positive bacteria, both have a thick cell wall, which may have implications on their digestibility (Salinas et al., 2007; Yu et al., 2015). In sum, nematode food choice was determined by bacterial traits (e.g. morphology and secondary metabolite production), rather than morphological traits of the nematode grazers (e.g. size of buccal cavity).

4.2. Pumping activity of bacterial grazers

The pharyngeal pumping activity was mainly determined by the nematode genus and pharynx morphology, i.e. three-part (*Diploscapter*) and two-part pharynx (*Plectus*). Particularly, the relation of pharynx to body length differed between *Diploscapter* (19.6 ± 2.5 %) and *Plectus* (22.7 ± 3.2 %; $p < 0.001$). While the mean pumping frequency across bacterial species was nearly identical, with 63 and 62 pumps min^{-1} for *Plectus* and *Diploscapter*, respectively, the pumping patterns varied between both taxa: *Plectus* showed high pumping rates for *E. coli* OP50 and low rates for *B. subtilis* and *Kitasatospora* sp., but the opposite was observed for *Diploscapter*. The pumping activity is generally linked to food-density, i.e. the pumping frequency decreases with less bacterial food source available (Chiang et al., 2006; Lee et al., 2017; Fueser et al., 2021). Moreover, *Plectus* as general opportunist is adapted to lower food densities compared to the enrichment opportunist *Diploscapter,* which in turn can affect the regulation of the pumping frequency (Woombs and Laybourn-Parry, 1984).

Contrary to our expectations, the preference for a bacterial species in the choice assay was not related to an increase in pumping rate. For example, *E. coli* OP50 was preferred by *Diploscapter* in most choice assays, but triggered a downregulation of the pumping frequency, even though *Diploscapter* was cultivated on *E. coli* OP50. This further contrasts Song et al. (2013), reporting that the exposure to a familiar food increased the pumping frequency of *C. elegans*. The present results indicate that a positive relationship between common food bacteria and pharyngeal pumping activity cannot be generalized across taxa. In line with that, Liu et al. (2018) stated that the familiarity of a diet is less important for soil-dwelling nematodes while foraging. In detail, *Mesorhabditis* and *Acrobeloides* were reared for 6 months on a bacterial species to develop food familiarity. Consecutive choice assays with novel bacterial taxa as alternative diet revealed that food preference was not affected by familiarity but correlated with food quality, i.e. the food that best supports growth and reproduction (Liu et al., 2018).

Interestingly, *Legionella* distinctly enhanced the pumping rates of both *Diploscapter* species compared to *E. coli* OP50, although it was avoided in the choice assay. Presumably, the assay time was too short for the pathogen to colonize the digestive tract, establish LCVs and impair nematode fitness. The intestinal colonization of *Legionella* in *C. elegans* was around 10^2 and 10^4 CFU per individual after 24 h and 48 h, respectively (Brassinga et al., 2010). In line with this, avoidance of *Legionella* in the choice assay was first observed after 8 h and increased within 48 h. Similarly, avoidance of *B. subtilis* and *Kitasatospora* sp. by *Diploscapter* in the food choice assay was not related to a decrease in pumping rate. Further, a strong preference for *P. fluorescens* over *E. coli* OP50 by *Plectus* was not correlated to an increase of pumping frequencies compared to the latter. Taken together, food quality (i.e. beneficial or detrimental for the nematode grazer) apparently is of minor importance for the regulation of pumping frequency.

Another factor to consider in pharyngeal pumping assays is, that although activity is an indicator of food ingestion (Abada et al., 2009; Scholz et al., 2017; Lee et al., 2017), it is not necessarily related to its digestion. For instance, under harmful UV-light exposure *C. elegans* reverses its muscle contraction patterns, thereby producing a spitting behavior that ejects ingested material from the pharynx (Bhatla et al., 2015; Sando et al., 2021). Moreover, the present study revealed that bacterial size had no effect on pumping activity. This was unexpected, as small cells are ingested more efficiently than large ones (Avery and Shtonda, 2003). Other traits affecting the pumping rate may be the viscosity and water content of the bacterial colonies. Suzuki et al. (2020) revealed that the higher the viscosity of an *E. coli* suspension, the lower the pharyngeal pumping rate of *C. elegans*. However, in the current study all bacterial species were suspended in PBS, thus the viscosity of the offered bacterial strains was comparable. Further, *C. elegans* showed similar pumping rates when fed with polystyrene beads or *E. coli* at a density of 10^7 particles ml^{-1}, indicating that water content has no effect on pumping rates at lower food densities (Fueser et al., 2021).

Overall, these results suggest that the pumping frequency alone is no suitable indicator for food quality or even food choice, especially since bacterial traits such as species and size were less relevant factors. In contrast to food choice, the pumping activity was modulated by nematode traits such as genus and pharynx morphology.

4.3. Foraging of nematodes

The current results revealed that food preference and food ingestion are independent processes, which describe different components of the nematode foraging strategy. The latter is shaped by several complex processes including prey searching, recognition, consumption and digestion (Jeschke et al., 2002), determining the functional response of a predator (Chan et al., 2017). In nematode food choice assays the driving factors are firstly the search for the prey via chemotaxis, i.e. detecting bacterial metabolites (Kim and Flavell, 2020). Once a bacterial prey is encountered, bacterial traits are regulating the food preference. Typically, a prey is chosen, that is easily captured, handled and ingested with low energetic costs (Hohberg and Traunspurger, 2005; Liu et al., 2018). This was revealed by the choice assay, as the CI of the different nematode taxa was significantly related to bacterial size and species.

On the other hand, morphological adaptions of bacteria increase the handling time required for capturing, consuming and digestion (Jousset, 2012). Generally, in bacterial grazers such as nematodes, the relationship between ingestion and prey density is described by a type III functional response, i.e. with increasing food density the consumption rate follows a sigmoidal shape (Holling, 1959; Ptatscheck, 2021). In detail, the higher the bacterial density, the higher the amount of bacteria ingested. This relationship applies until a maximum bacterial density is reached. Above this maximum, the consumption rate stagnates because it is then limited by the handling time, i.e. the amount of bacteria that can be ingested per time unit.

By using similar bacterial cell numbers, the present study showed that if density is no factor, the pumping rate is related to nematode morphology, but not to bacterial size and species. This indicates that once specific bacteria are selected, bacterial traits are no longer crucial for ingestion and nematode grazer traits gain importance.

The foraging strategy of nematodes must also be considered in relation to the soil food web. As shown in the pharyngeal pumping assay, nematodes are able to increase their ingestion rate when confronted with a less preferred bacterial diet as sole food source, which may be an

advantage when competition is high and food density is low. Generalist feeding in microbial grazers is a strategy also observed for other soil fauna, e.g. collembolans and mites (Sechi et al., 2014; Erktan et al., 2020). Moreover, this implies that nematodes also promote nutrient cycling when the preferred food is not available. Overall, this dietary flexibility contributes to the stability and function of the soil food web.

5. Conclusion

The food choice and pharyngeal pumping activity of nematodes are two independent processes that shape nematode feeding behavior. While the food choice is mainly determined by bacterial traits, the pumping rate is more related to nematode traits. Thus, the pharyngeal pumping assay is no suitable alternative approach to investigate nematode food preference. Rather, both assays should be conducted in parallel to disentangle the different components of the nematode foraging strategy.

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CRediT authorship contribution statement

Christin Ortlieb: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Ben Katzer:** Methodology, Investigation. **Robin Liess:** Methodology, Investigation. **Martha Gartz:** Methodology, Investigation. **Michael Steinert:** Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Liliane Ruess:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.pedobi.2024.151018.

Data Availability

Data will be made available on request.

References

Abada, E.A.E., Sung, H., Dwivedi, M., Park, B.J., Lee, S.K., Ahnn, J., 2009. *C. elegans* behavior of preference choice on bacterial food. Mol. Cells 28, 209–213.
Aitlhadj, L., Stürzenbaum, S.R., 2013. The toxicological assessment of two anti-obesity drugs in *C. elegans*. Toxicol. Res. 2, 145–150.
Altun, Z.F., Hall, D.H., 2009. Alimentary System, Pharynx. In WormAtlas. doi:10.3908/wormatlas.1.3.
Avery, L., Shtonda, B.B., 2003. Food transport in the *C. elegans* pharynx. J. Exp. Biol. 206, 2441–2457.
Avery, L., Bargmann, C.I., Horvitz, H.R., 1993. The *Caenorhabditis elegans unc-31* gene affects multiple nervous system-controlled functions. Genetics 134, 455–464.
Bargmann, C.I., Hartwieg, E., Horvitz, H.R., 1993. Odorant-selective genes and neurons mediate olfaction in *C. elegans*. Cell 74, 515–527.
Ben Arous, J., Laffont, S., Chatenay, D., 2009. Molecular and sensory basis of a food related two-state behavior in *C. elegans*. PLoS One 4, e7584.
Berini, F., Katz, C., Gruzdev, N., Casartelli, M., Tettamanti, G., Marinelli, F., 2018. Microbial and viral chitinases: Attractive biopesticides for integrated pest management. Biotechnol. Adv. 36, 818–838.
Bhatla, N., Droste, R., Sando, S.R., Huang, A., Horvitz, H.R., 2015. Distinct neural circuits control rhythm inhibition and spitting by the myogenic pharynx of *C. elegans*. Curr. Biol. 25, 2075–2089.
Bongers, T., 1990. The maturity index: an ecological measure of environmental disturbance based on nematode species composition. Oecologia 83, 14–19.
Bongers, T., Ferris, H., 1999. Nematode community structure as a bioindicator in environmental monitoring. Trends Ecol. Evol. 14, 224–228.
Bonkowski, M., Villenave, C., Griffiths, B., 2009. Rhizosphere fauna: the functional and structural diversity of intimate interactions of soil fauna with plant roots. Plant Soil 321, 213–233.
Brassinga, A.K.C., Kinchen, J.M., Cupp, M.E., Day, S.R., Hoffman, P.S., Sifri, C.D., 2010. *Caenorhabditis* is a metazoan host for *Legionella*. Cell. Microbiol. 12, 343–361.
de Carvalho, M.J.A., Mirth, C.K., 2017. Food intake and food choice are altered by the developmental transition at critical weight in *Drosophila melanogaster*. Anim. Behav. 126, 195–208.
Chan, K., Boutin, S., Hossie, T.J., Krebs, C.J., O'Donoghue, M., Murray, D.L., 2017. Improving the assessment of predator functional responses by considering alternate prey and predator interactions. Ecology 98, 1787–1796.
Chiang, J.T.A., Steciuk, M., Shtonda, B., Avery, L., 2006. Evolution of pharyngeal behaviors and neuronal functions in free-living soil nematodes. J. Exp. Biol. 209, 1859–1873.
van den Hoogen, J., Geisen, S., Routh, D., Ferris, H., Traunspurger, W., Wardle, D.A., de Goede, R.G.M., Adams, B.J., Ahmad, W., Andriuzzi, W.S., Bardgett, R.D., Bonkowski, M., Campos-Herrera, R., Cares, J.E., Caruso, T., de Brito Caixeta, L., Chen, X., Costa, S.R., Creamer, R., Mauro da Cunha Castro, J., Dam, M., Djigal, D., Escuer, M., Griffiths, B.S., Gutiérrez, C., Hohberg, K., Kalinkina, D., Kardol, P., Kergunteuil, A., Korthals, G., Krashevska, V., Kudrin, A.A., Li, Q., Liang, W., Magilton, M., Marais, M., Martín, J.A.R., Matveeva, E., Mayad, E.H., Mulder, C., Mullin, P., Neilson, R., Nguyen, T.A.D., Nielsen, U.N., Okada, H., Rius, J.E.P., Pan, K., Peneva, V., Pellissier, L., Carlos Pereira da Silva, J., Pitteloud, C., Powers, T. O., Powers, K., Quist, C.W., Rasmann, S., Moreno, S., Scheu, S., Setälä, H., Sushchuk, A., Tiunov, A.V., Trap, J., van der Putten, W., Vestergård, M., Villenave, C., Waeyenberge, L., Wall, D.H., Wilschut, R., Wright, D.G., Yang, J.I., Crowther, T.W., 2019. Soil nematode abundance and functional group composition at a global scale. Nature 572, 194–198.
Diomede, L., Rognoni, P., Lavatelli, F., Romeo, M., del Favero, E., Cantu, L., Ghibaudi, E., di Fonzo, A., Corbelli, A., Fiordaliso, F., Palladini, G., Valentini, V., Perfetti, V., Salmona, M., Merlini, G., 2014. A *Caenorhabditis elegans*–based assay recognizes immunoglobulin light chains causing heart amyloidosis. Blood, J. Am. Soc. Hematol. 123, 3543–3552.
Erktan, A., Or, D., Scheu, S., 2020. The physical structure of soil: determinant and consequence of trophic interactions. Soil Biol. Biochem. 148, 107876.
Fang-Yen, C., Avery, L., Samuel, A.D., 2009. Two size-selective mechanisms specifically trap bacteria-sized food particles in *Caenorhabditis elegans*. Proc. Natl. Acad. Sci. 106, 20093–20096.
Ferris, H., 2010. Contribution of nematodes to the structure and function of the soil food web. J. Nematol. 42, 63.
Flavell, S.W., Pokala, N., Macosko, E.Z., Albrecht, D.R., Larsch, J., Bargmann, C.I., 2013. Serotonin and the neuropeptide PDF initiate and extend opposing behavioral states in *C. elegans*. Cell 154, 1023–1035.
Fox, J., Weisberg, S., 2019. An R Companion to Applied Regression, Third edition. Sage, Thousand Oaks, California.
Fradin, H., Kiontke, K., Zegar, C., Gutwein, M., Lucas, J., Kovtun, M., Corcoran, D.L., Ryan Baugh, L., Fitch, D.H.A., Piano, F., Gunsalus, K.C., 2017. Genome architecture and evolution of a unichromosomal asexual nematode. Cur. Biol. 27, 2928–2939.
Franks, C.J., Holden-Dye, L., Bull, K., Luedtke, S., Walker, R.J., 2006. Anatomy, physiology and pharmacology of *Caenorhabditis elegans* pharynx: a model to define gene function in a simple neural system. Invertebr. Neurosci. 6, 105–122.
Fueser, H., Rauchschwalbe, M.T., Höss, S., Traunspurger, W., 2021. Food bacteria and synthetic microparticles of similar size influence pharyngeal pumping of *Caenorhabditis elegans*. Aquat. Toxicol. 235, 105827.
Gaur, A.V., Agarwal, R., 2021. Risperidone induced alterations in feeding and locomotion behavior of *Caenorhabditis elegans*. Cur. Res. Toxicol. 2, 367–374.
Gibbs, D.S., Anderson, G.L., Beuchat, L.R., Carta, L.K., Williams, P.L., 2005. Potential role of *Diploscapter* sp. strain LKC25, a bacterivorous nematode from soil, as a vector of food-borne pathogenic bacteria to preharvest fruits and vegetables. Appl. Environ. Microbiol. 71, 2433–2437.
Hellinga, J.R., Garduño, R.A., Kormish, J.D., Tanner, J.R., Khan, D., Buchko, K., Jimenez, C., Pinette, M.M., Brassinga, A.K.C., 2015. Identification of vacuoles containing extraintestinal differentiated forms of *Legionella pneumophila* in colonized *Caenorhabditis elegans* soil nematodes. MicrobiologyOpen 4, 660–681.
Hemmerling, C., Labrosse, A., Ruess, L., Steinert, M., 2023. *Legionella pneumophila* and Free-Living Nematodes: Environmental Co-Occurrence and Trophic Link. Microorganisms 11, 738.
Higaki, W.A., Araujo, F.F., 2012. *Bacillus subtilis* e abamectina no controle de nematoides e alterações fisiológicas em algodoeiro cultivado em solos naturalmente infestados. Nematropica 42, 295–303.
Hohberg, K., Traunspurger, W., 2005. Predator–prey interaction in soil food web: functional response, size-dependent foraging efficiency, and the influence of soil texture. Biol. Fertil. Soils 41, 419–427.
Holling, C.S., 1959. The components of predation as revealed by a study of small-mammal predation of the European pine sawfly. Can., Entomol. 91, 293–320.
Izquierdo, P.G., Charvet, C.L., Neveu, C., Green, A.C., Tattersall, J.E., Holden-Dye, L., O'connor, V., 2023. Modelling organophosphate intoxication in *C. elegans* highlights nicotinic acetylcholine receptor determinants that mitigate poisoning. PLoS One 18, e0284786.

Jeschke, J.M., Hohberg, K., 2008. Predicting and testing functional responses: an example from a tardigrade–nematode system. Basic Appl. Ecol. 9, 145–151.
Jeschke, J.M., Kopp, M., Tollrian, R., 2002. Predator functional responses: discriminating between handling and digesting prey. Ecol. Monogr. 72, 95–112.
Jeschke, J.M., Kopp, M., Tollrian, R., 2004. Consumer-food systems: why type I functional responses are exclusive to filter feeders. Biol. Rev. 79, 337–349.
Jousset, A., 2012. Ecological and evolutive implications of bacterial defences against predators. Environ. Microbiol. 14, 1830–1843.
Kassambara, A., 2023. rstatix: Pipe-Friendly Framework for Basic Statistical Tests. R package version 0.7.2. ⟨https://CRAN.R-project.org/package=rstatix⟩.
Kavitha, K., Mathiyazhagan, S., Sendhilvel, V., Nakkeeran, S., Chandrasekar, G., Dilantha Fernando, W.G., 2005. Broad spectrum action of phenazine against active and dormant Structures of fungal pathogens and root knot nematode. Arch. Phytopathol. Plant Prot. 38, 69–76.
Khan, F., Jain, S., Oloketuyi, S.F., 2018. Bacteria and bacterial products: Foe and friends to *Caenorhabditis elegans*. Microbiol. Res. 215, 102–113.
Kim, D.H., Flavell, S.W., 2020. Host-microbe interactions and the behavior of *Caenorhabditis elegans*. J. Neurogenet. 34, 500–509.
Kuznetsova, A., Brockhoff, P.B., Christensen, R.H.B., 2017. lmerTest Package: Tests in Linear Mixed Effects Models. J. Stat. Softw. 82, 1–26.
Laing, S.T., Ivens, A., Butler, V., Ravikumar, S.P., Laing, R., Woods, D.J., Gilleard, J.S., 2012. The transcriptional response of *Caenorhabditis elegans* to ivermectin exposure identifies novel genes involved in the response to reduced food intake. PloS One 7, e31367.
Lee, K.S., Iwanir, S., Kopito, R.B., Scholz, M., Calarco, J.A., Biron, D., Levine, E., 2017. Serotonin-dependent kinetics of feeding bursts underlie a graded response to food availability in *C. elegans*. Nat. Commun. 8, 14221.
Lewenza, S., Charron-Mazenod, L., Giroux, L., Zamponi, A.D., 2014. Feeding behavior of *Caenorhabditis elegans* is an indicator of *Pseudomonas aeruginosa* PAO1 virulence. PeerJ 2, e521.
Liu, T., Yu, L., Xu, J., Yan, X., Li, H., Whalen, J.K., Hu, F., 2017. Bacterial traits and quality contribute to the diet choice and survival of bacterial-feeding nematodes. Soil Biol. Biochem. 115, 467–474.
Liu, T., Yu, L., Li, M., Wu, J., Li, H., Whalen, J.K., Hu, F., 2018. Food familiarity does not change nematode feeding behavior. Soil Biol. Biochem. 125, 136–143.
Liu, T., Yu, F., Zhou, M., Xu, J., Wu, J., Hu, F., Li, H., Whalen, J.K., 2021. Soil fauna actively change their diet to survive stress. Soil Biol. Biochem. 162, 108435.
Luedtke, S., O'Connor, V., Holden-Dye, L., Walker, R.J., 2010. The regulation of feeding and metabolism in response to food deprivation in *Caenorhabditis elegans*. Invertebr. Neurosci. 10, 63–76.
Matz, C., Kjelleberg, S., 2005. Off the hook–how bacteria survive protozoan grazing. Trends Microbiol 13, 302–307.
Mazzuchelli, R.D.C.L., Mazzuchelli, E.H.L., de, Araujo, F.F., 2020. Efficiency of *Bacillus subtilis* for root-knot and lesion nematodes management in sugarcane. Biol. Control 143, 104185.
Meisel, J.D., Kim, D.H., 2014. Behavioral avoidance of pathogenic bacteria by *Caenorhabditis elegans*. Trends Immunol. 35, 465–470.
Moens, T., Vincx, M., 1997. Observations on the feeding ecology of estuarine nematodes. J. Mar. Biol. Assoc. U. K. 77, 211–227.
Moens, T., Yeates, G.W., De Ley, P., 2004. Use of carbon and energy sources by nematodes. Nematol. Monogr. Perspect. 529–545.
Molmeret, M., Horn, M., Wagner, M., Santic, M., Abu Kwaik, Y., 2005. Amoebae as training grounds for intracellular bacterial pathogens. Appl. Environ. Microbiol. 71, 20–28.
Monteiro, L., Brinke, M., dos Santos, G., Traunspurger, W., Moens, T., 2014. Effects of heavy metals on free-living nematodes: A multifaceted approach using growth, reproduction and behavioural assays. Eur. J. Soil Biol. 62, 1–7.
Newsham, K.K., Rolf, J., Pearce, D.A., Strachan, R.J., 2004. Differing preferences of Antarctic soil nematodes for microbial prey. Eur. J. Soil Biol. 40, 1–8.
Otoguro, K., Nakagawa, A., Omura, S., 1988. Setamycin, a 16-membered macrolide antibiotic identification and nematocidal activity. J. Antibiot. 41, 250–252.
Pacheco, P.V.M., Campos, V.P., Terra, W.C., Pedroso, M.P., de Paula, L.L., da Silva, M.S. G., Monteiro, T.S.A., de Freitas, L.G., 2022. Attraction and toxicity: Ways volatile organic compounds released by *Pochonia chlamydosporia* affect *Meloidogyne incognita*. Microbiol. Res. 255, 126925.
Paranjape, K., Bédard, É ., Shetty, D., Hu, M., Choon, F.C.P., Prévost, M., Faucher, S.P., 2020. Unravelling the importance of the eukaryotic and bacterial communities and their relationship with *Legionella* spp. ecology in cooling towers: a complex network. Microbiome 8, 1–19.
Pereira, T.H.R., de Moura, T.R., Santos, M.R.M., dos Santos Zamarioli, L., Erustes, A.G., Smaili, S.S., Pereira, G.J.S., de Godoy Netto, A.V., Bincoletto, C., 2024. Palladium (II) compounds containing oximes as promising antitumor agents for the treatment of osteosarcoma: An in vitro and in vivo comparative study with cisplatin. Eur. J. Med. Chem. 264, 116034.
Ptatscheck, C., 2021. Role of nematodes in the food web: nematodes as predator and prey. Ecol. Freshw. Nematodes 216, 216–246.
R Core Team, 2023. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. ⟨https://www.R-project.org/⟩.
Rao, M.S., Kamalnath, M., Umamaheswari, R., Rajinikanth, R., Prabu, P., Priti, K., Grace, G.N., Chaya, M.K., Gopalakrishnan, C., 2017. *Bacillus subtilis* IIHR BS-2 enriched vermicompost controls root knot nematode and soft rot disease complex in carrot. Sci. Hortic. 218, 56–62.
Rasch, J., Krüger, S., Fontvieille, D., Ünal, C.M., Michel, R., Labrosse, A., Steinert, M., 2016. *Legionella*-protozoa-nematode interactions in aquatic biofilms and influence of Mip on *Caenorhabditis elegans* colonization. Int. J. Med. Microbiol. 306, 443–451.
Rasmann, S., Ali, J.G., Helder, J., van der Putten, W.H., 2012. Ecology and evolution of soil nematode chemotaxis. J. Chem. Ecol. 38, 615–628.
Ray, A., Rentas, C., Caldwell, G.A., Caldwell, K.A., 2015. Phenazine derivatives cause proteotoxicity and stress in *C. elegans*. Neurosci. Lett. 584, 23–27.
Richter, A., Kern, T., Wolf, S., Struck, U., Ruess, L., 2019. Trophic and non-trophic interactions in binary links affect carbon flow in the soil micro-food web. Soil Biol. Biochem. 135, 239–247.
Rønn, R., Vestergård, M., Ekelund, F., 2012. Interactions between bacteria, protozoa and nematodes in soil. Acta Protozool. 51, 223–235.
Ruess, L., 2024. Nematodes and their trophic interactions in the soil microbiome. In: Dunfield, K.E. (Ed.), Understanding and utilising soil microbiomes for a more sustainable agriculture. Burleigh Dodds Science Publishing Limited, Cambridge, UK, pp. 1–29.
Salinas, K.A., Edenborn, S.L., Sexstone, A.J., Kotcon, J.B., 2007. Bacterial preferences of the bacterivorous soil nematode *Cephalobus brevicauda* (Cephalobidae): effect of bacterial type and size. Pedobiologia 51, 55–64.
Sando, S.R., Bhatla, N., Lee, E.L., Horvitz, H.R., 2021. An hourglass circuit motif transforms a motor program via subcellularly localized muscle calcium signaling and contraction. Elife 10, e59341.
Scholz, M., Dinner, A.R., Levine, E., Biron, D., 2017. Stochastic feeding dynamics arise from the need for information and energy. Proc. Natl. Acad. Sci. 114, 9261–9266.
Sechi, V., D'Annibale, A., Ambus, P., Sárossy, Z., Krogh, P.H., Eriksen, J., Holmstrup, M., 2014. Collembola feeding habits and niche specialization in agricultural grasslands of different composition. Soil Biol. Biochem. 74, 31–38.
Shtonda, B.B., Avery, L., 2006. Dietary choice behavior in *Caenorhabditis elegans*. J. Exp. Biol. 209, 89–102.
Song, B.M., Avery, L., 2013. The pharynx of the nematode *C. elegans*: A model system for the study of motor control. Worm 2, e21833.
Song, B.M., Faumont, S., Lockery, S., Avery, L., 2013. Recognition of familiar food activates feeding via an endocrine serotonin signal in *Caenorhabditis elegans*. Elife 2, e00329.
Stephens, D.W., Krebs, J.R., 1986. Foraging theory, first ed. Princeton University Press, U.K.
Stuhr, N.L., Curran, S.P., 2020. Bacterial diets differentially alter lifespan and healthspan trajectories in *C. elegans*. In: Commun. Biol., 3, p. 653.
Suzuki, Y., Kikuchi, K., Numayama-Tsuruta, K., Ishikawa, T., 2020. How do *Caenorhabditis elegans* worms survive in highly viscous habitats? J. Exp. Biol. 223, 224691.
Tahseen, Q., Sheridan, J., Perry, R., 2004. Electropharyngeograms of pharyngeal pumping activity in six species of free-living nematodes. Nematology 6, 49–54.
Tierney, A.J., 2020. Feeding, hunger, satiety and serotonin in invertebrates. Proc. R. Soc. B. 287, 20201386.
Venables, W.N., Ripley, B.D., 2002. Modern Applied Statistics with S, Fourth Edition. Springer, New York.
Venette, R.C., Ferris, H., 1998. Influence of bacterial type and density on population growth of bacterial-feeding nematodes. Soil Biol. Biochem. 30, 949–960.
von Lieven, A.F., 2003. Functional morphology and evolutionary origin of the three-part pharynx in nematodes. Zoology 106, 183–201.
Whalen, J.K., Kernecker, M.L., Thomas, B.W., Sachdeva, V., Ngosong, C., 2013. Soil food web controls on nitrogen mineralization are influences by agricultural practices in humid temperate climates. CAB Rev. Persp. Agric. Vet. Sci. Nutr. Nat. Res. 8, 1–18.
Wintermeyer, E., Ludwig, B., Steinert, M., Schmidt, B., Fischer, G., Hacker, J., 1995. Influence of site specifically altered Mip proteins on intracellular survival of *Legionella pneumophila* in eukaryotic cells. Infect. Immun. 63, 4576–4583.
Woombs, M., Laybourn-Parry, J., 1984. Feeding biology of *Diplogasteritus nudicapitatus* and *Rhabditis curvicaudata* (Nematoda) related to food concentration and temperature, in sewage treatment plants. Oecologia 64, 163–167.
Yeates, G.W., 2010. Nematodes in ecological webs. In: Encyclopedia of Life Sciences. John Wiley & Sons Ltd, Hoboken, pp. 1–11.
Yu, L., Yan, X., Ye, C., Zhao, H., Chen, X., Hu, F., Li, H., 2015. Bacterial respiration and growth rates affect the feeding preferences, brood size and lifespan of *Caenorhabditis elegans*. PloS One 10, e0134401.
Zell, H., 1993. Die Gattung *Plectus* Bastian, 1865 Sensu lato (Nematoda, Plectidae)–ein Beitrag zur Ökologie, Biogeographie, Phylogenie und Taxonomie der Plectidae, Staatliches Museum für Naturkunde Karlsruhe: Karlsruhe, Germany, pp. 1–172.
Zhang, J., Bakheet, R., Parhar, R.S., Huang, C.H., Hussain, M.M., Pan, X., Siddiqui, S.S., Hashmi, S., 2011. Regulation of fat storage and reproduction by Krüppel-like transcription factor KLF3 and fat-associated genes in *Caenorhabditis elegans*. J. Mol. Biol. 411, 537–553.
Zheng, J., Dini-Andreote, F., Luan, L., Geisen, S., Xue, J., Li, H., Sun, B., Jiang, Y., 2022. Nematode predation and competitive interactions affect microbe-mediated phosphorus dynamics. mBio 13, e0329321.
Zhou, Y., Falck, J.R., Rothe, M., Schunck, W.H., Menzel, R., 2015. Role of CYP eicosanoids in the regulation of pharyngeal pumping and food uptake in *Caenorhabditis elegans*. J. Lipid Res. 56, 2110–2123.